

Passive Model-Predictive Impedance Control for Safe Physical Human–Robot Interaction

Ran Cao^{id}, Long Cheng^{id}, *Senior Member, IEEE*, and Houcheng Li^{id}

Abstract—Various cognitive systems have been designed to model the position and stiffness profiles of human behavior and then to drive robots by mimicking the human’s behavior to accomplish physical human–robot interaction tasks through a properly designed impedance controller. However, some studies have shown that variable stiffness parameters of the impedance controller can cause the violation of the passivity constraint of the robot states, and make the robot’s stored energy exceed the external energy injected from the human user, thus leading to the unsafe human–robot interaction. To solve this problem, this article proposes a novel passive model-predictive impedance control method including two control loops. In the bottom-loop of the proposed controller, the robot is driven by a variable impedance controller to achieve the desired compliant interaction behavior. In the top-loop of the proposed controller, the model-predictive control (MPC) is used to ensure that the robot states satisfy the passivity constraint by calculating a complementary torque to limit the stored energy of the robot. The passivity of the closed-loop robot system and the feasibility of MPC are guaranteed by theoretical analysis, ensuring the safety of the robotic movement in the human–robot interaction. The effectiveness of the proposed method is demonstrated by the simulation and experiment on the Franka Emika Panda robot.

Index Terms—Impedance control, model-predictive control (MPC), passivity, physical human–robot interaction.

I. INTRODUCTION

IN RECENT years, robots have been widely applied in many physical human–robot interaction application scenarios, such as assisting the human operators in the manufacturing factory or helping the therapists conduct the rehabilitation training [1], [2]. In these scenes, to keep a compliant human–robot interaction, the impedance control has been proposed to regulate the dynamic relationship between the robotic motion

and the interaction force as a virtual mass-spring-damper system [3], [4], [5]. One main challenge in the impedance control is that the controller’s parameters are set to be constant which cannot adapt to different interaction requirements and unstructured environments [6].

Research on the human’s central nervous system has shown that human adjusts the arm’s mechanical stiffness to interact with unstable environments [7]. Motivated by this finding, the variable impedance controller integrated with a cognitive system has been proposed to complete more complex interaction tasks. The cognitive system generates the controller’s variable stiffness parameters according to the task rules or the movement model obtained from human demonstrations [8], [9]. For example, based on the task rule of the minimal assistance, the assist-as-needed control scheme was proposed to determine the stiffness parameters of the impedance controller according to the human intention [10]. When the robot needs to assist the human user, the stiffness parameter of the controller is set to be a high value. When the human user tries to move actively, the stiffness parameter is set to be a low value. Based on the stiffness model encoded from the human’s demonstrations, Roza et al. [11] and Abu-Dakka et al. [12] generated time-varying stiffness parameters for a variable impedance controller to drive the robot to reproduce a human-like interaction behavior. Though the above studies introduced how to design the variable stiffness profile to complete interaction tasks, it has been noted that variable stiffness parameters of the impedance controller may threaten the stability of the physical human–robot interaction [13].

Passivity theory provides an energy-based tool to evaluate the coupled stability of physical human–robot interaction [14], [15]. A robot system is passive if the stored energy of the robot is always lower than the external energy injected from the environment, which means that the robot does not generate the extra energy. Research has shown that the interaction between robot and human is stable when the robot system is passive [16]. Motivated by this idea, a virtual energy-storing tank strategy was designed to preserve the passivity of the robot under a variable impedance controller by storing the robot’s passivity violating energy [17]. The main shortcoming of the method based on the virtual energy-storing tank is that its performance depends on the manually designed initial tank energy. Bednarczyk et al. [18] designed a passivity filter to modify variable stiffness parameters when the robot system is not passive. However, the proposed filter requires the stiffness parameter matrix of the impedance controller to be diagonal, which limits its applications. In addition, the proposed filter exhibits excessive conservativeness when adjusting stiffness parameters based on passivity constraints,

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and therefore does not achieve the minimum correction of the system.

Some studies show that the brain has an internal model of the environment and its own body, and uses this model to compute the optimal control inputs that minimize a cost function that reflects the task goals and constraints [19], [20]. This process involves solving a dynamic optimization problem that balances the tradeoff between accuracy and effort. Model-predictive control (MPC) adopts a similar scheme to seek the optimal control performance with multiple constraints in an online way [21], [22], [23], [24], [25].

Recent papers have shown that MPC can be applied in interaction control. In [26], based on a predicted contact force between the robot arm and the environment, the authors used MPC to calculate the path points of the robot arm to interact with the environment. Matschek et al. [27] combined a direct force control strategy and a model-predictive tracking control method. The proposed approach drove the robot to track the path while desired contact forces can be maintained. In [28], an MPC framework for lower limb exoskeletons combined with a human intention identification strategy based on a fuzzy logic algorithm was proposed. The designed algorithm achieved an assist-as-needed control for patients. However, the above methods cannot guarantee a compliant interaction between the robot and the human user. To address this issue, Bednarczyk et al. [29] designed a model-predictive impedance controller by combining the MPC method with the impedance control. One essential difference between this article and [29] is that the impedance controller in [29] is only employed to generate a desired trajectory for the MPC. In addition, the model-predictive impedance controller cannot satisfy the passivity constraint that has a close relationship with the safety of the human-robot interaction.

To solve the above problems and achieve a safe human-robot interaction, this article proposes a passive model-predictive impedance control (PMPIC) to combine impedance control and MPC with a passivity guarantee. First, an impedance controller with variable stiffness parameters is applied to transform the robot's system model into a desired mass-spring-damper system model. Based on this impedance model, MPC calculates a feasible complementary input to make the robot states satisfy the passivity constraint. The optimization objective of MPC is to minimize the complementary input, which is employed to reduce the conservativeness of the controller. The passivity of the proposed controller with variable stiffness profiles is theoretically analyzed, and the feasibility of the MPC problem is guaranteed. The main contributions of this article consist of the following two parts.

- 1) The proposed PMPIC combines the advantages of variable impedance control and MPC. On the one hand, it can drive the robot to achieve a compliant interaction behavior with the human user. On the other hand, the passivity condition of the closed-loop human-robot system can be satisfied.
- 2) Compared with the existing passivity preserving strategies [18], the conservativeness of the proposed controller is lower due to the optimality of the MPC, which means that the modification to the original stiffness profiles

under the proposed controller is lower. Additionally, the proposed controller does not require to make any assumption on the controller's stiffness parameters.

II. METHODOLOGY

In this section, the PMPIC scheme consisting of the bottom-loop variable impedance control (Section II-A) and the top-loop MPC (Section II-B) is introduced. The entire structure of the proposed control framework is shown in Fig. 1.

A. Bottom-Loop: Variable Impedance Control

An n -joint robotic manipulator model operating in an m -dimensional task space is described in the following Euler-Lagrange form:

$$M(q(t))\ddot{q}(t) + C(q(t), \dot{q}(t))\dot{q}(t) + g(q(t)) = \tau_c(t) + J^T(q(t))f_{in}(t) \quad (1)$$

where $q(t), \dot{q}(t), \ddot{q}(t) \in \mathbb{R}^n$ denote the joint position, velocity, and acceleration, respectively, $M(q(t)) \in \mathbb{R}^{n \times n}$ is the symmetric positive definite inertia matrix, $C(q(t), \dot{q}(t))\dot{q}(t)$, and $\tau_c(t) \in \mathbb{R}^n$ denote the Coriolis torque, the gravitational torque, and the control torque, respectively, $J(q(t)) \in \mathbb{R}^{m \times n}$ denotes the robot Jacobian matrix, and $f_{in}(t) \in \mathbb{R}^m$ denotes the interaction torque in task space.

A typical control law $\tau_c(t) = M(q(t))v(t) + C(q(t), \dot{q}(t))\dot{q}(t) + G(q(t)) - J^T(t)f_{in}(t)$ is employed to linearize the manipulator model (1) into a double-integrator model

$$\ddot{q}(t) = v(t) \quad (2)$$

where $v(t)$ denotes a new control command in joint space. To obtain the system model in task space, by using a forward differential kinematic model $\ddot{p}(t) = J(q(t))\ddot{q}(t) + \dot{J}(q(t), \dot{q}(t))\dot{q}(t)$ where $p(t) \in \mathbb{R}^m$ is the robot end-effector position, the double-integrator model defined by (2) in joint space can be transformed into the task space form as follows:

$$\ddot{p}(t) = u_{task}(t) \quad (3)$$

where the control command $u_{task}(t)$ satisfies $v(t) = J^+(q(t))u_{task}(t) - J^+(q(t))\dot{J}(q(t), \dot{q}(t))\dot{q}(t)$ where $J^+(q(t))$ denotes the Moore-Penrose inverse of $J(q(t))$. In order to guarantee the robot can interact compliantly with a human user, impedance control is employed to make the robot behave like a virtual compliant mass-spring-damper system under the interaction force. A typical impedance control command $u_{task}(t)$ is designed as

$$u_{task}(t) = \ddot{p}_r(t) - M_v^{-1}[D_v(t)(\dot{p}(t) - \dot{p}_r(t)) + K_v(t)(p(t) - p_r(t)) - f_{in}(t)] \quad (4)$$

where $p_r(t) \in \mathbb{R}^m$ denotes the robot end-effector's reference position, $M_v \in \mathbb{R}^{m \times m}$ denotes the constant mass parameter, and $D_v(t) \in \mathbb{R}^{m \times m}$ and $K_v(t) \in \mathbb{R}^{m \times m}$ denote the time-varying damping and stiffness parameters of the virtual mass-spring-damper system, respectively. Define the tracking error $e(t) = p(t) - p_r(t)$. Under the control law (4), system (2) can be rewritten as follows:

$$M_v\ddot{e}(t) + D_v(t)\dot{e}(t) + K_v(t)e(t) = f_{in}(t). \quad (5)$$

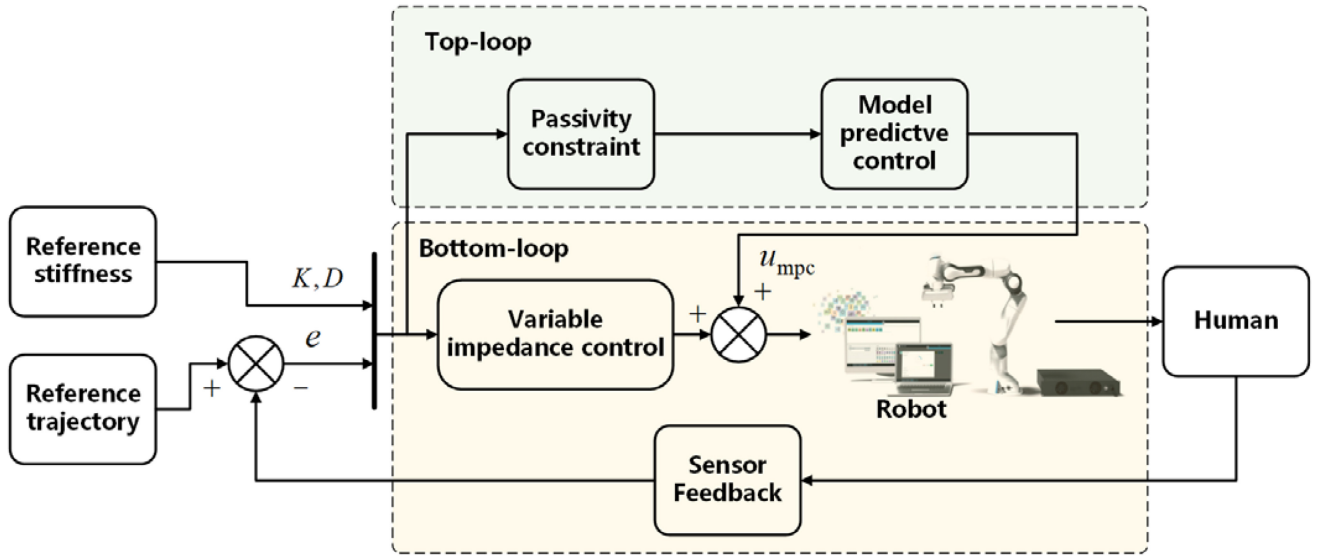


Fig. 1. Overview of the proposed PMPIC scheme.

Remark 1: In practice, the mass, damping, and stiffness parameters of the virtual mass-spring-damper system are determined according to the task requirement. Researchers design these parameter matrices based on practical experiences or the machine learning-based methods. However, some studies have pointed out that variable stiffness and damping parameters may threaten the entire closed-loop robot system's passivity feature that is necessary to achieve a stable interaction with the human user. In the following section, a passive model-predictive controller is proposed to guarantee the passivity of the entire system under varying stiffness and damping parameters.

B. Top-Loop: Passive Model-Predictive Control

First, the definition of passivity is given here. A general system model with the state variable $\chi(t) \in \mathbb{R}^n$, the input variable $u(t) \in \mathbb{R}^m$, and the output variable $y(t) \in \mathbb{R}^m$ can be written as the following state-space model:

$$\begin{aligned}\dot{\chi}(t) &= f(\chi(t), u(t)) \\ y(t) &= h(\chi(t), u(t)).\end{aligned}$$

The system is said to be passive if the input variable u and the output variable y satisfy the following condition with the initial state $\chi(0) \in \mathbb{R}^n$:

$$V(\chi(t)) - V(\chi(0)) \leq \int_0^t u^T(\tau) y(\tau) d\tau \quad (6)$$

where $V(\cdot)$ denotes a positive semidefinite energy storing function. The left hand side of (6) represents the stored energy, and the right hand side of (6) represents the external energy. In a human-robot interaction context, with a input signal $f_{in}(t)$ and a output signal $\alpha e(t) + \dot{e}(t)$ where α is a constant positive value and defined later, the passivity condition (6) ensures that the stored energy of the robot is always lower than the external energy injected from a human user. Under the assumption that a human user does not intentionally try to destabilize the interaction, the couple stability of the entire human-robot interaction system can be guaranteed [14].

The problem of how to make a variable impedance controller satisfy the passivity condition without a significant modification of the manually designed variable stiffness profile can be viewed as a constrained optimal control problem. In this article, MPC is employed to solve this finite-horizon optimal control problem with passivity constraint at each time step.

Based on the impedance controller designed in Section II-A, the closed-loop system model (5) integrated with an MPC output is

$$M_v \ddot{e}(t) + D_v(t) \dot{e}(t) + K_v(t) e(t) = f_{in}(t) + u_{mpc}(t). \quad (7)$$

The external energy in (6), i.e., the time integral of the inner product between an output signal $\alpha e(t) + \dot{e}(t)$ and a input signal $f_{in}(t)$ yields

$$\begin{aligned}& \int_0^t (\alpha e(\tau) + \dot{e}(\tau))^T f_{in}(\tau) d\tau \\&= \int_0^t (\alpha e(\tau) + \dot{e}(\tau))^T \left[M_v(\ddot{e}(\tau) + \alpha \dot{e}(\tau) - \alpha \dot{e}(\tau)) \right. \\&\quad \left. + D_v(\tau) \dot{e}(\tau) + K_v(\tau) e(\tau) - u_{mpc}(\tau) \right] d\tau \\&= \int_0^t \left[(\alpha e(\tau) + \dot{e}(\tau))^T M_v(\alpha \dot{e}(\tau) + \ddot{e}(\tau)) + e^T(\tau) C(\tau) \dot{e}(\tau) \right. \\&\quad \left. + \frac{1}{2} e^T(\tau) \dot{C}(\tau) e(\tau) \right] d\tau + \int_0^t \left[\dot{e}^T(\tau) (D_v(\tau) - \alpha M) \dot{e}(\tau) \right. \\&\quad \left. + e^T(\tau) (\alpha K_v(\tau) - \frac{1}{2} \dot{C}(\tau)) e(\tau) \right. \\&\quad \left. + (\alpha e(\tau) + \dot{e}(\tau))^T u_{mpc}(\tau) \right. \\&\quad \left. + e^T(\tau) (K_v(\tau) + \alpha D_v(\tau) - \alpha^2 M - C(\tau)) \dot{e}(\tau) \right] d\tau \quad (8)\end{aligned}$$

where $C(t)$ is a positive semidefinite matrix. To cancel out the cross-terms, let $C(t) = K_v(t) + \alpha D_v(t) - \alpha^2 M_v$. Note that

$$\begin{aligned}& \int_0^t (e^T(\tau) C(\tau) \dot{e}(\tau) + \frac{1}{2} e^T(\tau) \dot{C}(\tau) e(\tau)) d\tau \\&= \frac{1}{2} e^T(t) C(t) e(t) - \frac{1}{2} e^T(0) C(0) e(0).\end{aligned} \quad (9)$$

Taking (9) into (8), it has

$$\begin{aligned} & \int_0^t (\alpha e(\tau) + \dot{e}(\tau))^T f_{in}(\tau) d\tau \\ &= V(t) - V(0) \\ &+ \int_0^t (x^T(\tau)P(\tau)x(\tau) - x^T(\tau)Hu_{mpc}(\tau))d\tau \end{aligned} \quad (10)$$

where

$$\begin{aligned} V(t) &= \frac{1}{2}x^T(t) \begin{bmatrix} K_v(t) + \alpha D_v(t) & \alpha M \\ \alpha M & M \end{bmatrix} x(t) \\ P(t) &= \begin{bmatrix} \alpha K_v(t) - \frac{1}{2}(\dot{K}_v(t) + \alpha \dot{D}_v(t)) & 0_{m \times m} \\ 0_{m \times m} & D_v(t) - \alpha M_v \end{bmatrix} \\ x(t) &= \begin{bmatrix} e(t) \\ \dot{e}(t) \end{bmatrix}, \quad H = \begin{bmatrix} \alpha I_{m \times m} \\ I_{m \times m} \end{bmatrix}. \end{aligned}$$

To ensure $V(t)$ is positive semi-definite, choose α as the ratio between the smallest eigenvalue of $D_v(t)$ and the largest eigenvalue of M_v . Note that $V(t)$ is the stored energy function in (6) and $W(t) = \int_0^t (x^T(\tau)P(\tau)x(\tau) - x^T(\tau)Hu_{mpc}(\tau))d\tau$ is the dissipated energy function. If the dissipated energy function $W(t)$ is nonnegative, i.e., satisfies the following condition:

$$\int_0^t (x^T(\tau)P(\tau)x(\tau) - x^T(\tau)Hu_{mpc}(\tau))d\tau \geq 0 \quad (11)$$

then the passivity condition $\int_0^t (\alpha e(\tau) + \dot{e}(\tau))^T f_{in}(\tau) d\tau \geq V(t) - V(0)$ holds, and the closed-loop robot system (7) is passive.

Remark 2: In the existing studies, the passivity condition (11) does not contain the complementary input $u_{mpc}(t)$ introduced in this article and can be written as $\int_0^t x^T(\tau)P(\tau)x(\tau)d\tau \geq 0$. As a result, the passivity of the robot system is only determined by the positive definitiveness of the matrix $P(t)$ that is closely related to the desired damping $D_v(t)$ and stiffness $K_v(t)$. If the desired damping and stiffness parameters are learned from human demonstrations, it is hard to ensure the positive definitiveness of $P(t)$. To address this issue, this article uses an extra complementary input $u_{mpc}(t)$ to make the passivity condition always holds, even if $P(t)$ may not be positive semidefinite. To minimize its effect on the desired interaction behavior, the complementary input is calculated by employing MPC when the norm of the complementary input is chosen as the cost function of MPC.

The system model (7) can be transformed into a state-space formulation as follows:

$$\dot{x}(t) = A(t)x(t) + Bu_{mpc}(t) + Bf_{in}(t) \quad (12)$$

where

$$\begin{aligned} A(t) &= \begin{bmatrix} 0_{n \times n} & I_{n \times n} \\ -M_v^{-1}K_v(t) & -M_v^{-1}D_v(t) \end{bmatrix} \\ B &= \begin{bmatrix} 0_{n \times n} \\ M_v^{-1} \end{bmatrix}. \end{aligned}$$

Discretizing system (12), it gives

$$\begin{aligned} x(t+1) &= f_x(x(t), u_{mpc}(t)) \\ &= \hat{A}(t)x(t) + \hat{B}u(t) + \hat{B}f_{in}(t) \end{aligned} \quad (13)$$

where $\hat{A}(t) = A(t)\Delta t + I_{2n \times 2n}$, $\hat{B} = B\Delta t$, and Δt denotes the unit sample time.

The condition (11) can be included in the MPC formulation by integrating with a generalized state variable z as follows:

$$\begin{aligned} \dot{z}(t) &= x^T(t)P(t)x(t) - x^T(t)Hu_{mpc}(t) \\ z(t) &\geq 0, z(0) = 0. \end{aligned} \quad (14)$$

Discretizing the auxiliary system (14), it can be obtained that

$$\begin{aligned} z(t+1) &= z(t) + \Delta t[x^T(t)P(t)x(t) - x^T(t)Hu_{mpc}(t)] \\ z(t) &\geq 0, z(0) = 0. \end{aligned} \quad (15)$$

Then the MPC problem can be written into the following form:

$$\begin{aligned} \min_{u_{mpc}(k|t)} \quad & J = \sum_{k=0}^{N-1} u_{mpc}(k|t)^T R u_{mpc}(k|t) \\ \text{s.t.} \quad & x(k+1|t) = \hat{A}(k|t)x(k|t) + \hat{B}u_{mpc}(k|t) + \hat{B}f_{in} \\ & z(k+1|t) = z(k|t) + \Delta t[x^T(k|t)P(k|t)x(k|t) \\ & \quad + x^T(k|t)Hu_{mpc}(k|t)] \\ & z(k|t) \geq 0 \\ & z(N|t) \geq 0 \\ & k = 0, \dots, N-1 \end{aligned} \quad (16)$$

where N denotes the prediction horizon, k denotes the k th sample time, $x(k|t)$ denotes the predicted state of the system at time step k for the current time t , and R is a constant weight matrix. Note that the cost function J only includes one complementary input term, because the proposed controller focuses on utilizing a minimum complementary input to make the system satisfy the passivity condition. In this article, the passivity of the robot system with the proposed model-predictive impedance controller is guaranteed by theoretical analysis. The feasibility of the MPC problem means that there always exists one solution $u_{mpc}(t)$ satisfying the passivity condition. Hence, the feasibility of the MPC problem and the passivity of the robot system are equivalent.

The main theorem is given here.

Theorem 1: The MPC problem (16) is recursively feasible, and the input signal $f_{in}(t)$ and the output signal $\alpha e(t) + \dot{e}(t)$ of the closed-loop system (5) have a passive relationship.

The proof can be found in the Appendix.

Remark 3: In practice, consider the fact that the external energy injected from a human user is always bounded, the passivity condition (11) can be relaxed by replacing $z(t) \geq 0$ with $z(t) \geq z_{\min}$, where $z_{\min}$ is a nonpositive constant. Here, $z_{\min}$ denotes the minimum acceptable excess of the external energy injected from a human user over the kinetic energy of the robot, which is determined according to practical experiences. By choosing an appropriate $z_{\min}$, the conservativeness of the passivity condition can be reduced without threatening the safety of the human-robot interaction.

III. VERIFICATIONS BY SIMULATION AND EXPERIMENT

To demonstrate the effectiveness of the proposed method, two simulations on the Franka Emika Panda robot are presented based on Gazebo and ROS platforms and one

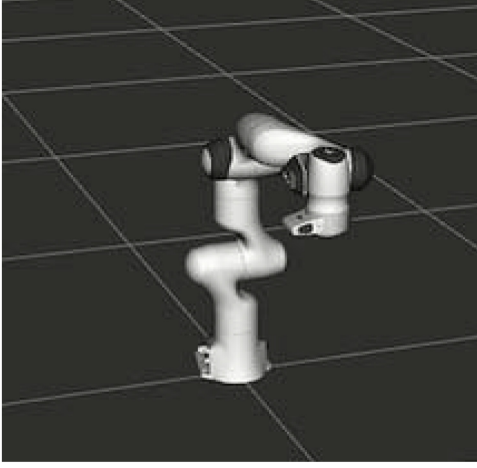


Fig. 2. Franka Emika Panda robot in simulations.

experiment is presented based on the physical Franka Emika Panda robot. In the first simulation, to show that the proposed method can guarantee the passivity of the variable impedance controller and the safety of the human–robot interaction, the proposed method is compared with one general variable impedance controller. In the second simulation, the passivity performance, referred to the dissipated energy and the modified stiffness value, are compared between the proposed controller and the passivity filter proposed in [18]. In the experiment, the effectiveness of the proposed controller is verified in a sawing task. The research on human subjects is reviewed and approved by the Ethics Committee of the Institute of Automation, Chinese Academy of Sciences (IA-201942), and the informed consent was signed by subjects.

In simulations and experiments, the dynamic model of Franka Emika Panda robot can be modeled using the Euler–Lagrange model based on (1). The inertia matrix, the Coriolis torque, and the gravitational torque are obtained from a C++ library called “Libfranka.” The impedance control at the bottom-loop of the whole controller runs at 1 kHz, and the MPC at the top-loop of the whole controller runs at 100 Hz. The solver of the optimization problem is chosen as OSQP [31] which has been broadly adopted because of its fast computation speed and desired performance. The generalized nonlinear system model is linearized at the current state to reduce the computation time.

A. Passivity Validation

In the first simulation, the control performance under the proposed model-predictive impedance controller satisfying the passivity constraint and the control performance under the general variable impedance controller without the passivity constraint are analyzed and compared. The virtual robot platform is shown in Fig. 2. The robot initially stays at the equilibrium position, and is then affected by the interaction force along y-axis after 3.5 s that satisfies

$$f_{in}(t) = \frac{5}{1 + e^{50(3.5-t)}} \text{ N.} \quad (17)$$

For the two controllers, the time-varying stiffness parameter of the impedance control is chosen as a 3-D diagonal matrix.

The three diagonal elements of the matrix k_x , k_y , and k_z denote three stiffness parameters of the robot along x-, y-, and z-axes, respectively. The varying stiffness parameter along y-axis is given by

$$k_y(t) = 22 + 12 \sin(t - \pi/2) \text{ N/m} \quad (18)$$

while the stiffness parameters along x- and z-axes are set as a constant value $k_x = k_z = 500 \text{ N/m}$. The mass matrix is set as $M_v = \text{diag}([1, 5, 1]) \text{ kg}$, and the damping ratio along y-axis is chosen as 0.01 to visualize the passivity effect while the damping ratios along x- and z-axes are chosen as 1. The compared impedance controller runs at 1 kHz, which is same as the running frequency of the proposed controller’s inner control loop.

The simulation results are presented in Fig. 3. The stiffness profile adopted in two controllers can be seen in Fig. 3(a). Fig. 3(b)–(d) show the position, velocity, and acceleration trajectories of the robot end-effector along y-axis under two controllers, respectively. Fig. 3(e) shows the dissipated energy $z(t)$ under two controllers. The complementary input $u_{mpc}(t)$ calculated in the proposed controller is shown in Fig. 3(f). Obviously, after the interaction force acted on the robot, the rapidly varying stiffness leads to the negative dissipated energy, which means that the stored energy exceeds the external energy produced by the interaction force. As a result, the general variable impedance controller cannot stabilize the robot under the interaction force, and the position, velocity, and acceleration trajectories of the robot end-effector gradually diverge. Meanwhile, the proposed controller calculates an appropriate complementary force to dissipate the extra energy, which ensures that the robot’s stored energy is always lower than the external energy. In conclusion, under the same parameter settings of the controllers, the proposed controller has the ability of making the robot’s states satisfy the passivity condition and stabilizing the robot while the robot states under the general impedance controller diverge.

B. Comparison Study

The second simulation is divided into two parts according to different comparison objectives. The first part demonstrates the influence of parameter z_{min} introduced in Remark 3, and the conservativeness of the proposed algorithm is analyzed. The second part shows the performance of a variable impedance controller with a passive filter proposed in [18], and its conservativeness is compared with the proposed controller. In the first part, the robot’s end-effector moves vertically and experiences a time-varying interaction force along y-axis that satisfies (17). The stiffness parameters along x-axis and z-axis are set as $k_x = 100 \text{ N/m}$ and $k_z = 100 \text{ N/m}$, respectively. Meanwhile, the stiffness parameter along y-axis is set as, $k_y = 10 + 500/(1 + e^{10(5.0-t)}) \text{ N/m}$. The damping parameters are determined to keep the damping ratio of 1 for the impedance model. The mass matrix is set as $M_v = \text{diag}([1, 5, 1]) \text{ kg}$. The prediction horizon for the MPC is set to $N = 10$ because a longer horizon requires more computing resources and more accurate system model, and a shorter horizon may affect the motion performance of

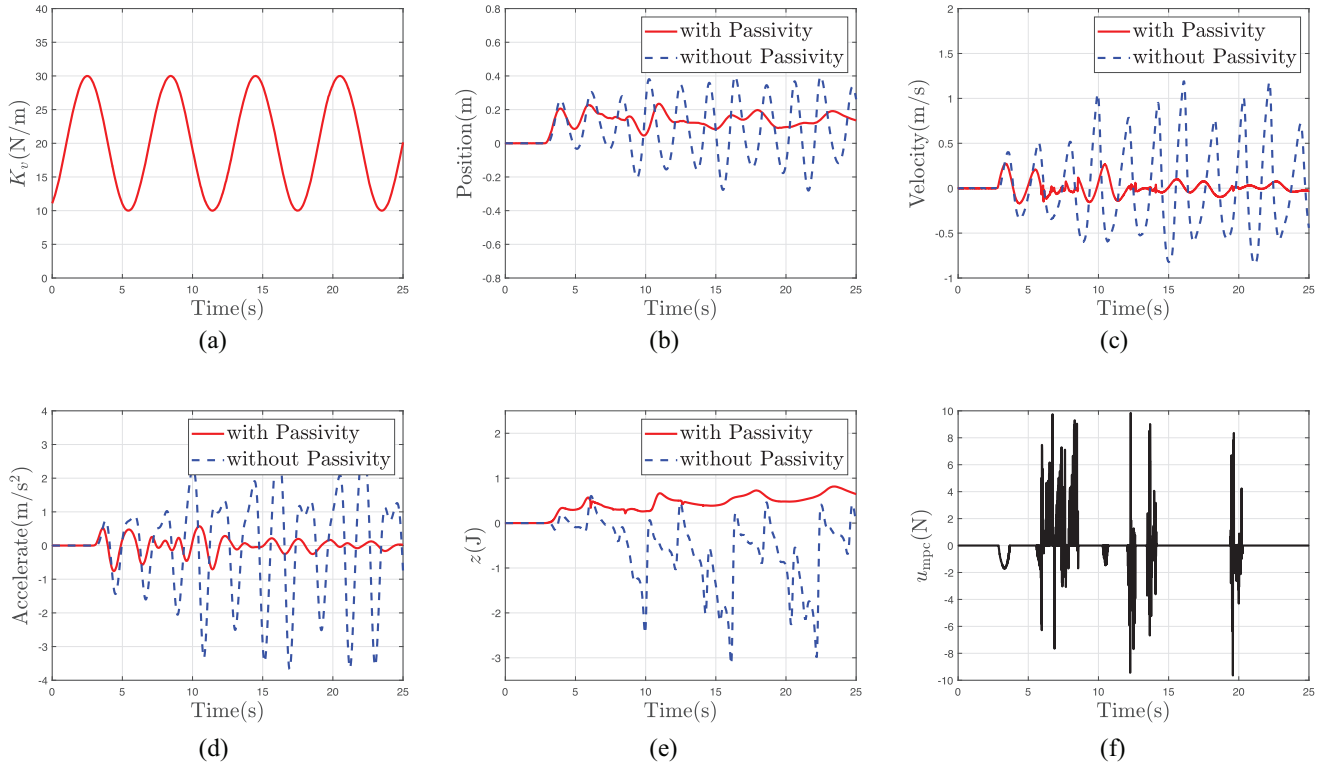


Fig. 3. Simulation results under the proposed PMPIC and the variable impedance controller without the passivity guarantee. (a) Variable stiffness profile. (b) Position trajectories. (c) Velocity trajectories. (d) Acceleration trajectories. (e) Dissipated energy profiles of the robot. (f) Complementary input.

the robot. The weight of the optimization target is set as $R = I_{3 \times 3}$. The first part of the simulation selects four groups of $z_{\min} = \{0.0, -1.0, -2.0, -3.0\}$, which means that the minimum acceptable excess of the external energy injected from a human user over the robot's kinetic energies are 0.0, 1.0, 2.0, and 3.0 J, respectively. In the second part, the compared variable impedance controller with a passivity filter works under the same task condition, stiffness matrix, damping matrix, and mass matrix. The filter parameters are chosen the same as the ones in [18]. Considering that the passivity filter maintains the passivity by modifying the stiffness parameters and damping parameters, one equivalent complementary input $u_{\text{mod}}(t)$ is calculated from the modification of the stiffness matrix $K_{\text{mod}}(t)$ and the damping matrix $D_{\text{mod}}(t)$ as follows:

$$u_{\text{mod}}(t) = K_{\text{mod}}(t)e(t) + D_{\text{mod}}(t)\dot{e}(t).$$

The conservativeness of two controllers can be judged by comparing the equivalent complementary input $u_{\text{mod}}(t)$ of the compared controller and the complementary input $u_{\text{mpc}}(t)$ of the proposed controller.

All simulation results are shown in Figs. 4 and 5. The meanings of different line types have been shown in legends, and the abbreviation “pf” denotes the compared variable impedance controller with the passivity filter. Fig. 4(a) demonstrates the time-varying stiffness parameter during the simulation. Fig. 4(b)–(d) indicates the position, velocity, and acceleration trajectories of the robot end-effector, respectively. The interaction force acting on the robot end-effector increases to 5N after $t = 3.5$ s and the complementary force $u_{\text{mpc}}(t)$ shown in Fig. 5 takes action to avoid the dissipated energy lower than

$z_{\min}$. The force combined with the interaction force and the complementary force leads to a drift in position, velocity, and acceleration trajectories. With a smaller $z_{\min}$, the lower bound of the dissipated energy of the robot is lower, and the chance that the passivity condition is violated is lower, which leads to a shorter time duration that the complementary force takes action and smoother position, velocity, and acceleration trajectories. Fig. 4(e) shows the dissipated energy of the robot that equal to the robot's external energy minus the stored energy injected from the interaction force. In the case that $z_{\min} = 0.0$, the minimum dissipated energy of $z(t)$ in the simulation is 0 J, which satisfies the passivity condition strictly. And in the case that $z_{\min} = \{-1.0, -2.0, -3.0\}$, the minimum dissipated energies of $z(t)$ in the simulation are -1.0 , -2.0 , and -3.0 J, respectively. Hence, it can be concluded that the proposed model-predictive impedance controller is effective. Meanwhile, as $z_{\min}$ gets closer to 0, the time that the complementary torque does not equal to 0 and takes effect is longer, and the maximum velocity and acceleration are lower, which means that the passivity condition is stricter and the interaction behavior is smoother. In conclusion, engineers can adjust $z_{\min}$ according to the detailed task requirement and safe limit. A lower $z_{\min}$ means that the modification to the original variable impedance behavior is less, however, it may lead to a loss of safety when the robot's stored energy is far greater than the external energy.

The simulation results under the variable impedance controller with the passivity filter indicate that the robot is also passive under the stiffness parameters modified by the passivity filter. However, as shown in the first subfigure of

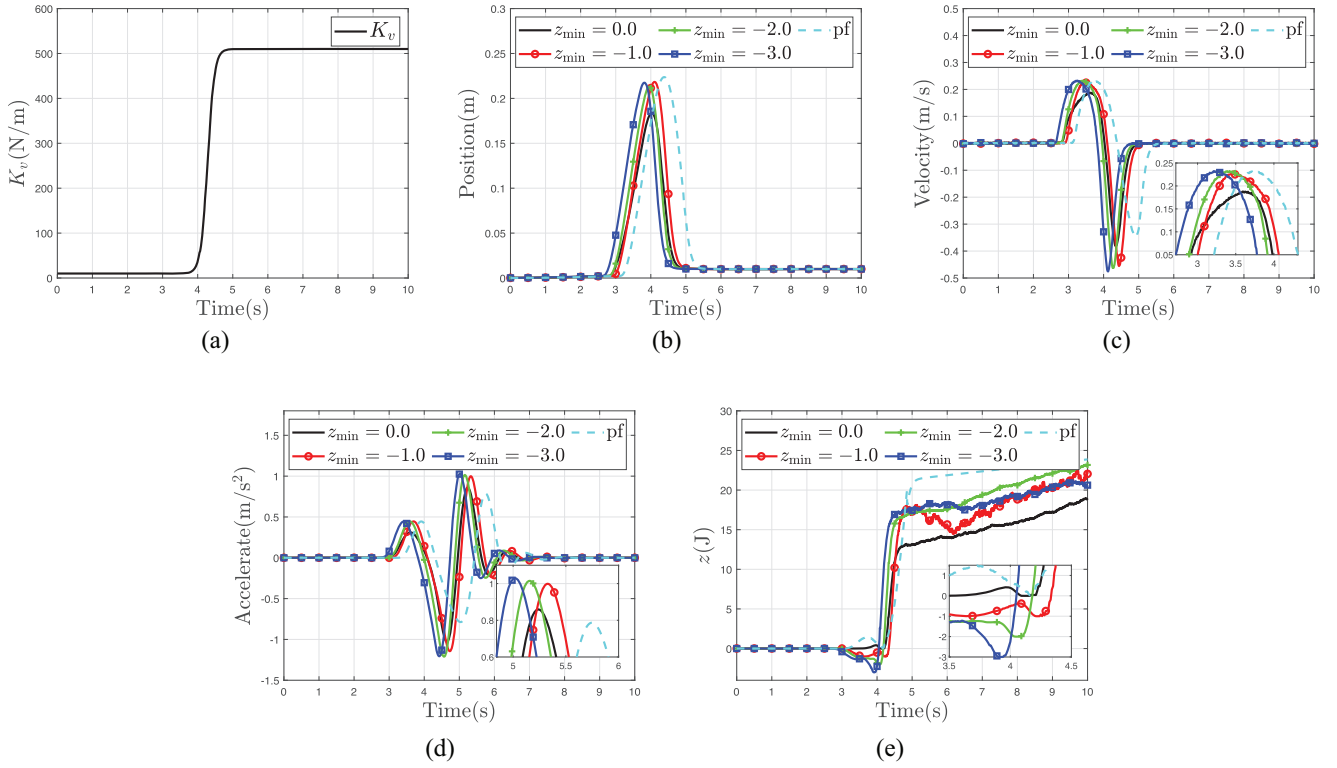


Fig. 4. Simulation results under the proposed PMPIC with four groups of $z_{\min}$ and the variable impedance controller with the passivity filter. (a) Variable stiffness profile. (b) Position trajectories. (c) Velocity trajectories. (d) Acceleration trajectories. (e) Dissipated energy profiles of the robot.

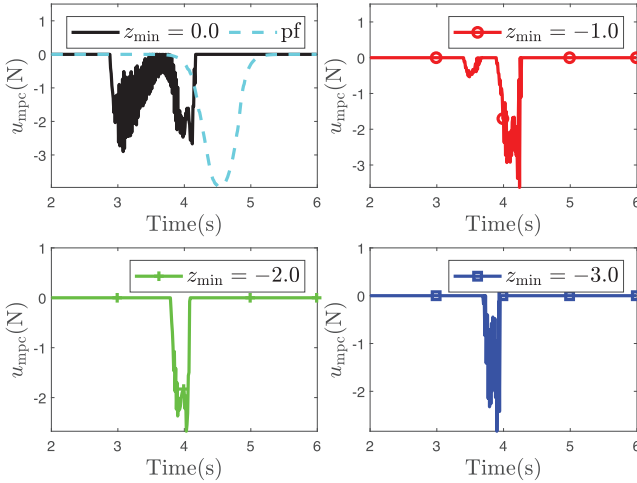


Fig. 5. Complementary input profiles under the proposed PMPIC with four groups of $z_{\min}$ and the variable impedance controller with the passivity filter.

Fig. 5, to achieve the passivity, the proposed controller with $z_{\min} = 0.0$ uses a lower complementary torque, and the maximum complementary torque is 26.92% lower than the maximum control torque calculated by the controller proposed in [18]. This means that the passive filter is more conservative than the proposed controller.

C. Physical Experiment

To validate the effectiveness of the proposed algorithm, one human-robot interaction experiment is conducted on the real Franka Emika Panda robot platform in this section.

In the experiment, the robot assists one human operator in completing a sawing task. The task consists of two stages, “push” and “pull.” In the stage of “push,” the robot maintains a low stiffness while following the human’s movement, and in the stage of “pull,” the robot keeps a high stiffness, forcing the person to move along with it. The entire duration of the experiment is 20 s, with two stages both lasting 10 s. During the experiment, two different disturbances were applied to the robot: 1) at 9.8 s, which is at the end of the first stage, the human operator tried to move the robot in the opposite direction and 2) at 15.2 s, which is in the middle of the second stage, another human applied the external force trying to stop the robot from moving. The snapshots of the experiment are shown in Fig. 6. The controller parameters are chosen as follows: the stiffness parameter along y-axis is set as $k_y = 30 + 40(1 + e^{2(10-t)})$ N/m, and the stiffness parameters along x-axis and z-axis are set as $k_x = k_z = 100$ N/m. In order to highlight the performance of the passivity-based control, this experiment sets the damping ratio along y-axis to be 0.8, and the damping ratios along x-axis and z-axis are still 1. The mass matrix is set as $M_v = \text{diag}([1, 1, 1])$ kg. The minimal acceptable excess of the robotic dissipated energy is $z_{\min} = -0.01$. The prediction horizon of MPC is 10. The weight matrix is $R = I_{3 \times 3}$.

The experimental results are shown in Fig. 7. Fig. 7(a)–(c) show the stiffness profile and the position trajectory, and the dissipated energy profile, respectively. It can be seen that under the use of the proposed PMPIC, the robot’s dissipated energy is greater than zero, which indicates that the robot’s kinetic energy is always lower than the external energy injected from the human.



Fig. 6. Snapshots of the sawing task.

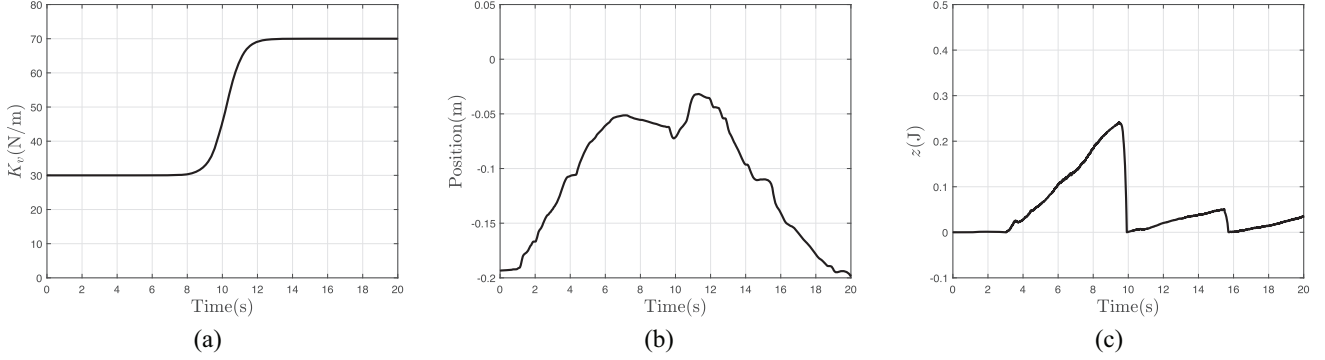


Fig. 7. Experiment results of the sawing task under the proposed PMPIC. (a) Variable stiffness profile. (b) Position trajectory. (c) Dissipated energy profile.

IV. DISCUSSION

Learning from demonstration enables robots to mimic the human's variable impedance skill by encoding varying human stiffness into a machine learning-based model. However, as shown in Section III-A, inappropriate stiffness and damping parameters can result in the loss of the passivity of the robot system, which leads to the unsafe human-robot interaction behavior. In this article, MPC supplies an efficient tool to preserve the passivity of the robot and to minimize the conservativeness of the controller while the stiffness and damping parameters are obtained from the machine learning based model. The comparison in Section III-B has verified this point. An obvious advantage brought by reducing the conservativeness is the variable impedance profile obtained from practical experience, manual rules, or machine learning methods can be preserved to a maximum extent. In addition, engineers can actively determine the conservativeness of the controller by adjusting $z_{\min}$. The main disadvantage of PMPIC is that the effectiveness of the MPC relies on an accurate robot dynamic model. This requirement can be relaxed partially by reducing the prediction horizon N in the MPC problem (16). One alternative method is to combine the existing robust MPC strategies with the passivity analysis. In future work, the authors plan to consider the uncertainty in the robot dynamics and to improve the robustness of PMPIC, and to reduce its dependence on the model accuracy.

V. CONCLUSION

This article provides an approach by combining the MPC and the impedance control to help the robot interact compliantly and safely with humans. The passivity condition of the model-predictive impedance control is satisfied, which ensures that robots complete the safe interaction task. The stiffness

parameters of the impedance control can be set as time-varying values, which allows the robot to adapt to different interaction task requirements. Meanwhile, the passivity of the robot can be guaranteed by the theoretical analysis, which means that the entire interaction is stable. Simulation results show that the maximum complementary input of the proposed controller is 26.92% lower than the maximum complementary input of the existing controller, which means that the proposed controller is less conservative and the variable stiffness feature is preserved to the some extent.

APPENDIX

PROOF OF THEOREM 1

Assume that $u^*(t) = \{u^*(0|t), \dots, u^*(N-1|t)\}$ is the optimal solution of the MPC problem (16) at time t , $x^*(t) = \{x^*(1|t), \dots, x^*(N|t)\}$ is the corresponding optimal state, and $z^*(t) = \{z^*(1|t), \dots, z^*(N|t)\}$ is the corresponding auxiliary system state. The following proof indicates that there exists a feasible solution at time $t+1$.

Consider the candidate solution

$$u_{\text{mpc}}(k|t+1) = \begin{cases} u^*(k+1|t), & k = 0, \dots, N-2 \\ u_s, & k = N-1 \end{cases} \quad (19)$$

where u_s is one solution satisfying the following feasibility condition:

$$\begin{aligned} & x^T(N-1|t+1)P(N-1|t+1)x(N-1|t+1) \\ & \geq x^T(N-1|t+1)Hu_s. \end{aligned} \quad (20)$$

To show that there exists at least one solution u_s , the situation that $x^T(N-1|t+1)P(N-1|t+1)x(N-1|t+1) = x^T(N-1|t+1)Hu_s$ is considered here. Note that: 1) if $x^T(N-1|t+1) = \mathbf{0}_{2m}$, u_s can be any real vector and 2) if $x^T(N-1|t+1) \neq \mathbf{0}_{2m}$, at least one solution u_s exists because $\text{rank}([x^T(N-1|t+1)P(N-1|t+1) - Hu_s]) > 0$.

$1|t+1)x(N-1|t+1) \mid x^T(n-1|t+1)H]) = \text{rank}([x^T(N-1|t+1)P(N-1|t+1)x(N-1|t+1)]) = 1$. Hence, u_s always exists.

Then at time $t+1$, $z(N|t+1)$ can be calculated as follows:

$$\begin{aligned} & z(N|t+1) \\ &= z(N-1|t+1) + \Delta t[-x^T(N-1|t+1)Hu_s \\ &\quad + x^T(N-1|t+1)P(N-1|t+1)x(N-1|t+1)] \\ &\geq z(N-1|t+1) \end{aligned} \quad (21)$$

where the second inequality is obtained by (20).

Substituting (19) into (13) and (15), the system state can be solved as follows:

$$x(k|t+1) = \begin{cases} x^*(k+1|t), & k = 1, \dots, N-1 \\ f_x(x^*(N|t), u_s), & k = N \end{cases} \quad (22)$$

$$z(k|t+1) = \begin{cases} z^*(k+1|t), & k = 1, \dots, N-1 \\ z(N|t+1), & k = N. \end{cases} \quad (23)$$

According to (21) and (23), it can be concluded that $z(N|t+1) \geq z(N-1|t+1) = z^*(N|t) \geq 0$. From (23), it is obvious that $z(k|t+1)$ satisfies the constraint $z(k|t+1) \geq 0$, $k = 0, \dots, N-1$. Hence, the MPC problem (16) is recursively feasible, and the closed-loop system (13) is always passive.

Remark 4: In some papers focusing on the passive control for the human-robot interaction, the output signal of the closed-loop system is chosen as $\dot{e}(t)$ rather than $\alpha e(t) + \dot{e}(t)$ like in this article. The main reason of the difference is that choosing $\dot{e}(t)$ as the output signal results in that the auxiliary complementary input u_{mpc} cannot always ensure the passivity of the closed-loop system. In this situation, the feasibility condition (20) can be rewritten as $\dot{e}^T(N-1|t+1)D_v(N-1|t+1)\dot{e}(N-1|t+1) - \frac{1}{2}e^T(N-1|t+1)\dot{K}_v(N-1|t+1)e(N-1|t+1) \geq \dot{e}^T(N-1|t+1)u_{\text{mpc}}(N-1|t+1)$. Obviously, it is impossible to find a suitable $u_{\text{mpc}}(N-1|t+1)$ to hold this passivity condition as $\dot{e}(N-1|t+1) = 0$, $e(N-1|t+1) \neq 0$, and $\dot{K}_v(N-1|t+1)$ is negative definite. If $\alpha e(t) + \dot{e}(t)$ is chosen as the output variable, there always exists $u_{\text{mpc}}(N-1|t+1)$ to satisfy the passivity condition (11).

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